Matias Liu Schmid

(843) 801-6997 · mliuschmid@gmail.com · matiasls.com · github.com/MatiasLiuS

EDUCATION

The George Washington University — B.S. Computer Science — Washington, DC · May 2025

Coursework: Probability & Statistics (Engineering Analysis III), Linear Algebra, Calculus I–II, Algorithms, Database Systems & Team Projects, Artificial Intelligence, Operating Systems, Discrete Structures

EXPERIENCE

TennisPlayer.net — Marketing & Automation Engineer (Independent Contractor) — Jun 2026 – Present

- Audited and restructured a live database of 170,000+ contact records, resolving conflicting membership-status flags and incomplete date fields, and set data-governance rules with the engineering lead.
- Built self-maintaining, rule-based segments (active, lapsed <365 days, leads, billing dunning/grace period) driven by date logic, eliminating manual list upkeep across a six-figure contact base.
- Compared reporting across WordPress and CRM analytics to define a unified measurement strategy spanning Google Analytics, Amazon SES deliverability data, and PostHog engagement metrics.
- Designed a campaign-naming taxonomy and tagging schema adopted as the data standard across the CRM; architected six automated customer journeys with time-delayed triggers and conditional branching.

Bart and Associates (BNA) — Software / IT Engineering Intern — Aug 2024 – Dec 2025

- Contributed to an AI job-matching prototype that scraped job postings across BNA's business ventures and matched them against current employees to surface internal-mobility and retention opportunities.
- Designed and built landing pages for BNA and several affiliated ventures, and owned end-to-end production of the company-wide annual newsletter.
- Remediated security vulnerabilities and managed patching alongside the IT lead; restored and stabilized a legacy WordPress site supporting business operations.

Thaddeus (nonprofit) — Web Development Intern — Aug 2024 – Dec 2024

- Built frontend features in React and Vite for a production platform within a full Agile/Scrum cycle; wrote end-to-end Playwright tests integrated into the CI/CD pipeline.

The George Washington University — Undergraduate Teaching Assistant — 2023 – 2025

- TA for Database Systems & Team Projects, Software Engineering, and AR/VR; held 4–6 hours of weekly office hours.
- Reviewed student schemas and ER diagrams for normalization problems, and helped untangle slow multi-table JOIN queries, Cartesian products, and injection-vulnerable SQL.

Tricorp — Lead Software & Technology Engineer — Aug 2021 – May 2023

- Sole technical owner of the company's digital infrastructure: took the business from zero online presence to a fully operational website built from scratch, and administered domains, DNS, hosting, and Microsoft 365.

PROJECTS

GESTURE — Wearable Sign-Language Translation (Senior Capstone, 5-person team) · *Python, PyTorch, Hugging Face, HDF5* — Aug 2024 – May 2025

- Owned the translation subsystem end to end: built a seq2seq Transformer from scratch in PyTorch to translate German Sign Language glosses into German text, then English via Meta's NLLB-200.
- Ran the bulk of hyperparameter tuning (learning rate, batch size, model dimensions, epochs), tracking BLEU-score improvements across iterations; debugged gradient-accumulation bottlenecks on long sequences.
- Rebuilt data loading, vocabularies, and tokenization when the project pivoted datasets; team pipeline used HDF5 storage with on-the-fly loading, padding, and batching.

ARGS — Academic Records & Graduation System (3-person team) · *Python, Flask, MySQL on AWS RDS* — Jan 2023 – May 2023

- Built the advising module: credit-weighted GPA computation, credit-hour and grade-threshold rules, and automated MS/PhD degree audits run against live transcript data in a 3NF/BCNF MySQL schema.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java, C, C#, HTML/CSS
Data & Analytics: MySQL, SQLite, AWS RDS, HDF5, schema design & normalization, CRM data segmentation, Google Analytics, PostHog
Web: React, Vite, Flask, Jinja, WordPress, Playwright, CI/CD
ML & Tooling: PyTorch, Transformers, Hugging Face, Git/GitHub, Linux